SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I ____V.Lasya(192524186)________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Answer:

1. Propositional Logic Representation:

Let R = It is raining, and W = The ground is wet.

Statement 1: $R \rightarrow W$

Statement 2: R

Goal: W

2. CNF Conversion:

$R \rightarrow W$ is equivalent to $\neg R \lor W$.

Therefore, the CNF clauses are: C1 = $(\neg R \lor W)$, C2 = (R).

3. Negate the Goal:

Goal = W

Negated goal = $\neg W$

So, C3 = ($\neg W$).

4. Resolution Algorithm:

Resolve C1 = ($\neg R \lor W$) with C2 = (R):

$(\neg R \lor W) \lor (R) \Rightarrow W$

Now resolve W with the negated goal $\neg W$:

$(W) \lor (\neg W) \Rightarrow \square$

5. Conclusion:

The Empty Clause (□) is derived. Therefore, the goal W is proved. Hence, the ground is wet.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Answer:

1. Propositional Logic Representation:

Let S = Rahul submitted the assignment, and M = Rahul receives internal marks.

Statement 1: $S \rightarrow M$

Statement 2: S

Goal: M

2. CNF Conversion:

$S \rightarrow M$ is equivalent to $\neg S \vee M$.

Therefore, C1 = $(\neg S \vee M)$, C2 = (S).

3. Negate the Goal:

Goal = M

Negated goal = $\neg M$

So, C3 = $(\neg M)$.

4. Resolution Algorithm:

Resolve C1 = $(\neg S \vee M)$ with C2 = (S):

$(\neg S \vee M) \vee (S) \Rightarrow M$

Now resolve M with the negated goal $\neg M$:

$(M) \vee (\neg M) \Rightarrow \square$

5. Conclusion:

The Empty Clause (□) is derived. Therefore, Rahul receives internal marks is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Answer:

1. Propositional Logic Representation:

Let L = Priya is a library member, and B = Priya can borrow books.

Statement 1: $L \rightarrow B$

Statement 2: L

Goal: B

2. CNF Conversion:

$L \rightarrow B$ is equivalent to $\neg L \vee B$.

Therefore, $C1 = (\neg L \vee B)$, $C2 = (L)$.

3. Negate the Goal:

Goal = B

Negated goal = $\neg B$

So, $C3 = (\neg B)$.

4. Resolution Algorithm:

Resolve $C1 = (\neg L \vee B)$ with $C2 = (L)$:

$(\neg L \vee B) \vee (L) \Rightarrow B$

Now resolve B with the negated goal $\neg B$:

$(B) \vee (\neg B) \Rightarrow \square$

5. Final Conclusion:

The Empty Clause ($\square$) is obtained. Therefore, Priya can borrow books is proved.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Answer:

1. Propositional Logic Representation:

Let A = Arun cleared the aptitude test, and E = Arun is eligible for placement.

Statement 1: $A \rightarrow E$

Statement 2: A

Goal: E

2. CNF Clauses:

$A \rightarrow E$ is equivalent to $\neg A \lor E$.

Therefore, C1 = $(\neg A \lor E)$, C2 = (A).

3. Negate the Goal:

Goal = E

Negated goal = $\neg E$

So, C3 = $(\neg E)$.

4. Resolution Algorithm:

Resolve C1 = $(\neg A \lor E)$ with C2 = (A):

$(\neg A \lor E) \lor (A) \Rightarrow E$

Now resolve E with the negated goal $\neg E$:

$(E) \lor (\neg E) \Rightarrow \square$

5. Resolution Tree:

```
(¬A ∨ E)        (A)
      \        /
        (E)
         |
       (¬E)
         |
          □
```

6. Conclusion:

The Empty Clause (□) is derived. Therefore, Arun is eligible for placement is proved.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.

2. If a user is authenticated, then the user is granted access.

3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

Answer:

1. Propositional Logic Representation:

Let P = The user entered the correct password, H = The user is authenticated, and G = The user is granted access.

Statement 1: $P \rightarrow H$

Statement 2: $H \rightarrow G$

Statement 3: P

Goal: G

2. CNF Conversion:

$P \rightarrow H$ is equivalent to $\neg P \lor H$.

$H \rightarrow G$ is equivalent to $\neg H \lor G$.

Therefore, the CNF clauses are: C1 = $(\neg P \lor H)$, C2 = $(\neg H \lor G)$, C3 = (P).

3. Negate the Goal:

Goal = G

Negated goal = $\neg G$

So, C4 = $(\neg G)$.

4. Resolution Algorithm:

First, resolve C1 = $(\neg P \lor H)$ with C3 = (P):

$(\neg P \lor H) \lor (P) \Rightarrow H$

Next, resolve H with C2 = $(\neg H \lor G)$:

$(H) \lor (\neg H \lor G) \Rightarrow G$

Finally, resolve G with the negated goal $\neg G$:

$(G) \lor (\neg G) \Rightarrow \square$

5. Final Conclusion:

The Empty Clause (□) is derived. Therefore, the user is granted access is proved.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation